

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	S Lakshmi priya	Reg. No.:	192524284
Section / Batch:	AI&DS	Date:	19/08/2026

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.
2. A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.
3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.
4. A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory-access bottlenecks, and increase computational throughput.
5. A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.

Code of Conduct

I S Lakshmi Priya(192524284)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: s priya

1. Memory Hierarchy for E-Commerce Platform

During peak shopping periods, thousands of users access product catalogs, customer records, and transaction data simultaneously. A proper memory hierarchy reduces latency and improves system performance.

Roles of Memory

Memory	Speed	Main Use
Cache Memory	Very High	Frequently accessed data and instructions
DRAM	High	Active applications, database buffers, sessions
Secondary Storage (SSD)	Lower	Permanent storage of products, customers, and transactions

Proposed Hierarchy

CPU



L1/L2/L3 Cache



DRAM



In-Memory Cache



NVMe SSD

- Cache: Stores frequently used data and instructions, providing very fast access.
- DRAM: Stores active application data, database indexes, and frequently accessed records.
- SSD: Stores persistent data such as customer records, orders, product databases, and backups.

Improving Performance	
Technique	Benefit
Caching popular products	Reduces repeated database access
Database buffering in DRAM	Reduces SSD access
Temporal locality	Keeps recently used data available
Spatial locality	Efficiently accesses nearby data
NVMe SSDs	Faster persistent read/write operations

Conclusion

A cache → DRAM → SSD hierarchy provides low-latency access while maintaining large storage capacity. Frequently accessed data stays in faster memory, while permanent transaction data remains on SSDs. This reduces access latency and supports high concurrent transaction throughput during peak shopping periods.

2. Memory Hierarchy for Robotic Manufacturing System

A robotic manufacturing system requires very low and predictable memory access times because camera processing, sensor readings, and control instructions must meet strict real-time deadlines.

Memory Suitability

Memory	Suitability	Use
SRAM	Excellent	Frequently accessed control data, cache
DRAM	Good	Camera images, sensor data, working memory
Non-Volatile Memory	Good	Program code, configuration, persistent data

Proposed Hierarchy

CPU



SRAM Cache



DRAM



Non-Volatile Memory

- SRAM: Stores critical control instructions and frequently accessed data because it provides very fast and predictable access.
- DRAM: Stores larger sensor readings, camera frames, and temporary processing data.
- Non-Volatile Memory: Stores program code, configuration, and data that must be retained after power loss.

Memory Allocation

Data	Memory	Reason
Control instructions	SRAM	Minimum latency
Real-time sensor data	SRAM + DRAM	Fast access with larger capacity
Camera images	DRAM	Large data size
Program code	Non-Volatile Memory + SRAM cache	Persistent and fast execution
Configuration	Non-Volatile Memory	Retained after power loss

Conclusion

A SRAM → DRAM → Non-Volatile Memory hierarchy provides both speed and capacity. Critical control data is kept in SRAM for predictable low-latency access, while DRAM handles larger sensor and camera data. Non-volatile memory stores programs and persistent information. This ensures the robot can meet strict real-time deadlines reliably.

3 Memory Hierarchy for Video-Editing Workstation

Processing multiple 4K/8K video streams requires high memory capacity and bandwidth. A suitable memory hierarchy can reduce delays and provide smooth real-time editing.

Factors Affecting Performance

Factor	Effect
Cache Size	Larger cache keeps frequently used data close to the CPU
DRAM Capacity	Holds large video frames and editing applications
Memory Bandwidth	Higher bandwidth transfers video data faster
Storage Performance	Fast SSDs reduce loading and file-transfer delays

Recommended Hierarchy

CPU/GPU



L1/L2/L3 Cache



High-Bandwidth DRAM



NVMe SSD



Large-Capacity Storage

- Cache: Stores frequently accessed processing data.
- DRAM: Provides large, fast working space for 4K/8K frames and multiple video streams.
- NVMe SSD: Provides fast loading, saving, and media streaming.
- Large storage: Stores completed projects and raw footage.

Performance Improvements

Technique	Benefit
Larger cache	Reduces CPU data-access delays
High-capacity DRAM	Prevents excessive swapping
High memory bandwidth	Faster video-frame transfer
NVMe SSD	Faster media loading and saving
Data locality	Reduces repeated memory access

Conclusion

A Cache → High-Bandwidth DRAM → NVMe SSD → Large Storage hierarchy provides fast data transfer and sufficient capacity. Larger DRAM and high memory bandwidth are especially important for handling multiple 4K/8K streams, enabling smooth real-time video editing with reduced processing delays.

4. Memory Hierarchy for Weather Forecasting System

A weather forecasting system processes huge datasets repeatedly. Efficient cache usage, high DRAM bandwidth, and large memory capacity can improve computational speed.

Factors Affecting Performance

Factor	Effect
Cache Hierarchy	Keeps frequently reused data close to the CPU
DRAM Bandwidth	Enables faster transfer of large datasets
Large-Capacity Memory	Holds massive atmospheric datasets without frequent storage access

Recommended Memory Hierarchy

CPU



L1/L2/L3 Cache



High-Bandwidth DRAM



Large-Capacity Memory



Secondary Storage

- Cache: Stores frequently reused temperature, pressure, and other data.
- DRAM: Holds active datasets and intermediate numerical calculations.
- Large-capacity memory: Allows large portions of the dataset to remain available for repeated computations.
- Secondary storage: Used for long-term datasets and results.

Improvements

Enhancement	Benefit
Larger cache	Reduces repeated memory accesses
High DRAM bandwidth	Faster dataset transfer
Data locality	Keeps related atmospheric data together
Large memory capacity	Reduces swapping/storage access
Blocking/tiling	Improves cache reuse

Conclusion

A Cache → High-Bandwidth DRAM → Large-Capacity Memory → Storage hierarchy improves data locality and reduces memory bottlenecks. Keeping frequently reused data in cache and active datasets in DRAM increases computational throughput and speeds up repeated weather-model calculations.

5 Memory Hierarchy for Smart Home Gateway

A smart home gateway continuously handles camera data, sensor readings, and appliance information. Since it has limited energy, the memory system should provide fast access while minimizing power consumption.

Memory Comparison

Memory	Latency	Capacity	Persistence	Power
SRAM	Very Low	Small	Volatile	High
DRAM	Low	Medium/Large	Volatile	Moderate
Flash	Higher	Large	Non-volatile	Low
EEPROM	Higher	Small	Non-volatile	Low

Proposed Memory Hierarchy

CPU



SRAM Cache



Low-Power DRAM

↓
Flash Storage

↓
Cloud/External Backup

- SRAM: Stores frequently accessed sensor values and control instructions for rapid access.
- DRAM: Stores active camera frames, sensor data, and running applications.
- Flash: Stores configuration, logs, device settings, and long-term data permanently.
- Cloud/External Storage: Used for optional backups and large historical data.

Energy Optimization

Technique	Benefit
Cache frequently used data	Reduces repeated DRAM/Flash access
Low-power DRAM	Reduces energy consumption
Flash for long-term storage	Provides persistent, low-power storage
Sleep modes	Turns off unused memory components

Conclusion

A SRAM → DRAM → Flash hierarchy provides fast access to frequently used data while keeping energy consumption low. SRAM handles critical data, DRAM manages active workloads, and Flash provides reliable long-term storage.

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 25	/ 15	/ 10	/ 5	/ 100

Course Coordinator

Faculty Incharge